

TEST REPORT

TRE00213A

A. Specifications

Model	PG-350		
HGS No.	V7MTGCCV		
Measuring Range	NO _x	25/50/100/250/500/1000/2500 ppm	
	SO ₂	200/500/1000/3000 ppm	
	CO	200/500/1000/2000/5000 ppm	
	CO ₂	10/20/30 vol%	
	O ₂	10/25 vol%	
Power	AC 100~240V 50/60Hz		
Output	DC 4~20mA·LAN		
MAC ADDRESS	00 05 2B 08 B1 E4		

B. Test Results

1. Appearance and Construction ----- Good

2. Function and Operation ----- Good

3. Performance (Note: %FS means percentage of full scale.)

1) Repeatability ----- Good

Parameter	NOx	SO ₂	CO	CO ₂	O ₂	
Test range	25 ppm	200 ppm	200 ppm	10 vol%	10 vol%	
Deviation	-0.1 %FS	0.4 %FS	0.0 %FS	-0.2 %FS	0.1 %FS	
Criteria	within ±1.0%FS (within ±0.5%FS for higher conc. than 1000ppm of CO and 100ppm of NOx)					

2) Linearity ----- Good

Parameter	NOx	SO ₂	CO	CO ₂	O ₂	
Test range	2500 ppm	3000 ppm	5000 ppm	30 vol%	25 vol%	
Deviation	-1.2 %FS	0.8 %FS	0.2 %FS	0.3 %FS	-0.4 %FS	
Criteria	within ±2.0%FS					

3) Drift ----- Good

Parameter	NOx	SO ₂	CO	CO ₂	O ₂	
Test range	25 ppm	200 ppm	200 ppm	10 vol%	10 vol%	
Zero (/day)	0.3 %FS	0.4 %FS	-0.1 %FS	-0.5 %FS	-0.4 %FS	
Criteria	within ±1.0%FS/day (within ±2.0%FS/day for SO ₂)					
Span (/day)	-0.4 %FS	0.0 %FS	0.3 %FS	0.2 %FS	0.1 %FS	
Criteria	within ±1.0%FS/day (within ±2.0%FS/day for SO ₂)					

4) Response Time ----- Good

Parameter	NOx	SO ₂	CO	CO ₂	O ₂	
Test range	25 ppm	200 ppm	200 ppm	10 vol%	10 vol%	
Td	24 s	20 s	20 s	19 s	21 s	
T90	10 s	16 s	11 s	11 s	13 s	
Td+T90	34 s	36 s	31 s	30 s	34 s	
Criteria(Td+T90)	smaller than 45s from sample inlet (smaller than 180s for SO ₂)					

5) Noise level (less than 1.0%FS p-p) ----- Good

6) Interference with coexisting gas ----- Good
(within ±2.0%FS, however higher conc. range than 200ppm for CO is within ±1.0%FS, and CH₄ interference on SO₂ measurement is within ±5.0%FS)

7) Supply Voltage Fluctuation ----- Good

Parameter	NOx	SO ₂	CO	CO ₂	O ₂	
Test range	25 ppm	200 ppm	200 ppm	10 vol%	10 vol%	
Influence	0.1 %FS	0.8 %FS	-0.1 %FS	0.1 %FS	0.0 %FS	
Criteria	within ±1.0%FS by ±10% fluctuation from rating voltage					

4. Withstand voltage (AC1400V between AC and E terminals for 5 seconds) ----- Good

5. Insulation Resistance(more than 5MΩ with DC1000V) ----- 2000 MΩ ----- Good

6. Leakage(less than 500Pa/5min. with 15kPa pressure) ----- -400 Pa ----- Good

7. NOx converter efficiency(more than 95%) ----- 95.8 % ----- Good

C. Overall Judgment ----- Pass

Temperature 25 °C Humidity 61 %RH

HORIBA, Ltd.

Date : 19 May 2015

Inspected by : *M. Fukai*

Approved by : *T. Toyonaka*